Jin Woo Oh

ojw925@gmail.com • 201-753-0265 • Atlanta, GA 30340 • linkedin.com/in/jinwoooh • ojw92.github.io

SKILLS

Languages & Frameworks: Python, SQL, R, C++, React, TypeScript, JavaScript, Zero, Flask, Ollama, CUDA, RAPIDS

Machine Learning & AI: Claude, Modal, Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-Learn (sklearn), HuggingFace, LangChain, fine-tuning, hypothesis testing, regression, anomaly detection, deep learning

Tools & DevOps: GitHub Actions, Docker, Kubernetes (k8s), AWS, GCP, Dagster, Temporal, Convex (NoSQL), PostgreSQL, Apache, Power BI, BigQuery, MCP, test automation (vitest, Playwright, pytest), Bash

EXPERIENCE

Avnir **Nov 2025 – Present**
Software Engineer *Atlanta, GA*

- Developed GeoPin, a location intelligence feature (self-hosted Nominatim on k8s) that recommends nearby contacts by goal-skill fit and finds low-resistance warm introduction path via Apache AGE graph on PostgreSQL; built the Python backend and React map UI, exposed over MCP for in-house AI agents
- Built Claude-powered contact-extraction pipeline, parallelizing large uploads by decomposing Temporal activity into concurrent batches, reducing ingestion time from 40 minutes to 4
- Created an end-to-end relationship-strength scoring model (SOM) with 30 behavioral features tuned via sklearn grid search, stratified k-fold CV, and teacher-student score distillation; cut pipeline reruns from 75 minutes to 5 seconds via per-user caching
- Automated Customer Success reporting pipeline by replacing a manual Google Sheet with 20-dataset/30-chart Apache Superset dashboard computing KPIs (DAU/WAU, engagement-depth, etc), deployed across 3 environments

ICC Lab, Florida State University **May 2025 – Nov 2025**
Researcher *Tallahassee, FL (Remote)*

- Automated 1,000+ hours of manual labeling by building a gaze-tracking pipeline with Tobii eye-tracking data and OpenCV to overlay real-time gaze positions and regions of interest on screen recordings to streamline behavioral analysis
- Achieved 95% accuracy in detecting UI state change via robust template matching and HSV color masking

FlightAware **June 2024 – Aug 2024**
Software Engineer Intern *Houston, TX (Remote)*

- Reduced latency of deep learning web application by 35% by developing scalable API (k8s, Streamlit & FastAPI) to provide secure internal access to the web service across multiple teams and earning positive client feedback
- Enhanced robustness and improved system performance of the Foresight Labs website by 20% via streamlining REST API calls tested on Postman, reconfiguring dependencies on Redis and Triton servers
- Performed A/B test of GPT-4o vs Mistral 7B RAG chatbots; recommended GPT-4o for 21x speed & 40% higher success

Samsung Electronics America **Aug 2019 – Apr 2024**
Software Quality Engineer III *Berkeley Heights, NJ*

- Spearheaded the development of a GPU-accelerated technical classifier model to identify customer complaints citing critical issues of flagship models with over 92% accuracy, automating 11+ hours of work per week
- Built and automated ETL pipeline (Python + SQL) processing 1M+ articles for downstream data classification and daily QA reporting of Android OS (9-14), improving data consistency and accelerating issue detection workflows by 25%
- Reduced operation cost of tech support live chat by 30% through a resource optimization simulation on Arena
- Presented clear, actionable business insights to stakeholders with user behavior & QA trend dashboards using Power BI

PROJECTS

Ahab | *pytest, SQLite, Qwen/Gemma, llama.cpp, FastAPI, Tailscale, Slack SDK, launchd, API, CUSUM, TSMOM, XGBoost*

- Autonomous nightly equity-analysis engine applying flow signals, self-hosted LLM interpretation, and Slack alert

RayTracing | *C++, CUDA, linear algebra, computer graphics, parallel programming, GPU acceleration, multi-threading, HPC*

- Implemented CUDA ray tracer achieving 35x GPU speedup over CPU via parallel programming and HPC scaling

Technical Issue Classifier | *PyTorch, TensorFlow, Scikit-Learn, HuggingFace, CUDA, BeautifulSoup, Docker, k8s, SQL*

- Optimized model by grid searching and threshold-tuning the ROC curve, minimizing costly false negatives to 3%

EDUCATION

Georgia Institute of Technology **Expected Dec 2026**
Master of Science (M.S.) in Analytics *Atlanta, GA (Online)*

Seoul National University **Feb 2018**
Master of Science (M.S.) in Mechanical Engineering *Seoul, Korea*

- 2 patents, 2 journal publications, 8 conference presentations; Academic Scholarship; 3.75/4.0 GPA, *Summa cum laude*

University of Illinois at Urbana-Champaign **May 2015**
Bachelor of Science (B.S.) in Mechanical Engineering *Champaign, IL*

- 1 journal publication; VP, Co-founder, mentor and tutor at Korean Mechanical Engineer Network; Academic Scholarship